

Generative Modeling

More on GANs and VAEs

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Laboratory for methods of big data analysis



LAMBDA • HSE

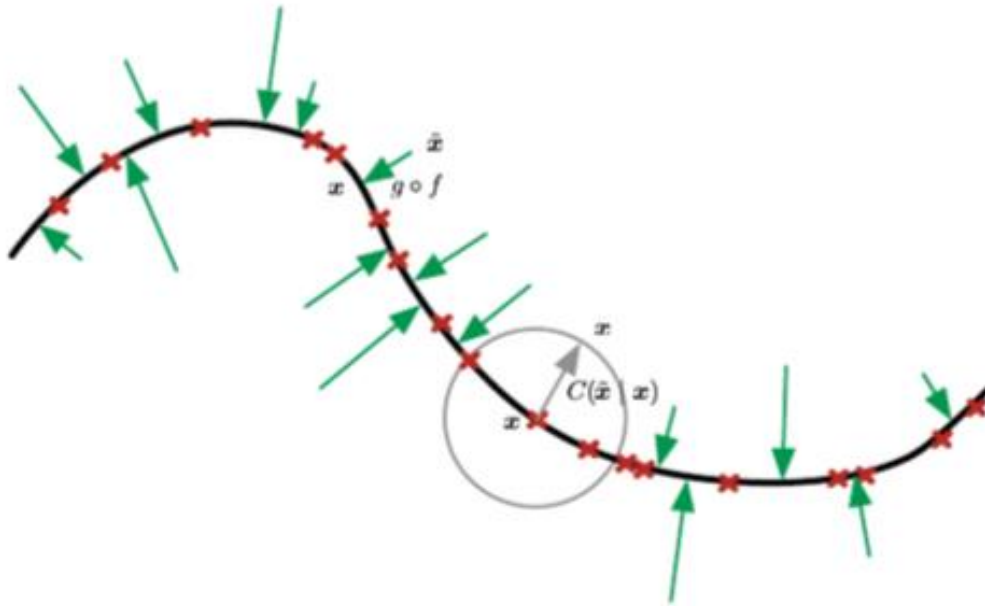
Spring 2025

In this Lecture

- ▶ Special Discriminator Structures
 - Energy-based Generative Adversarial Network
 - Boundary Equilibrium Generative Adversarial Networks
 - Discriminator Rejection Sampling
- ▶ Additional tips for the infrastructure.

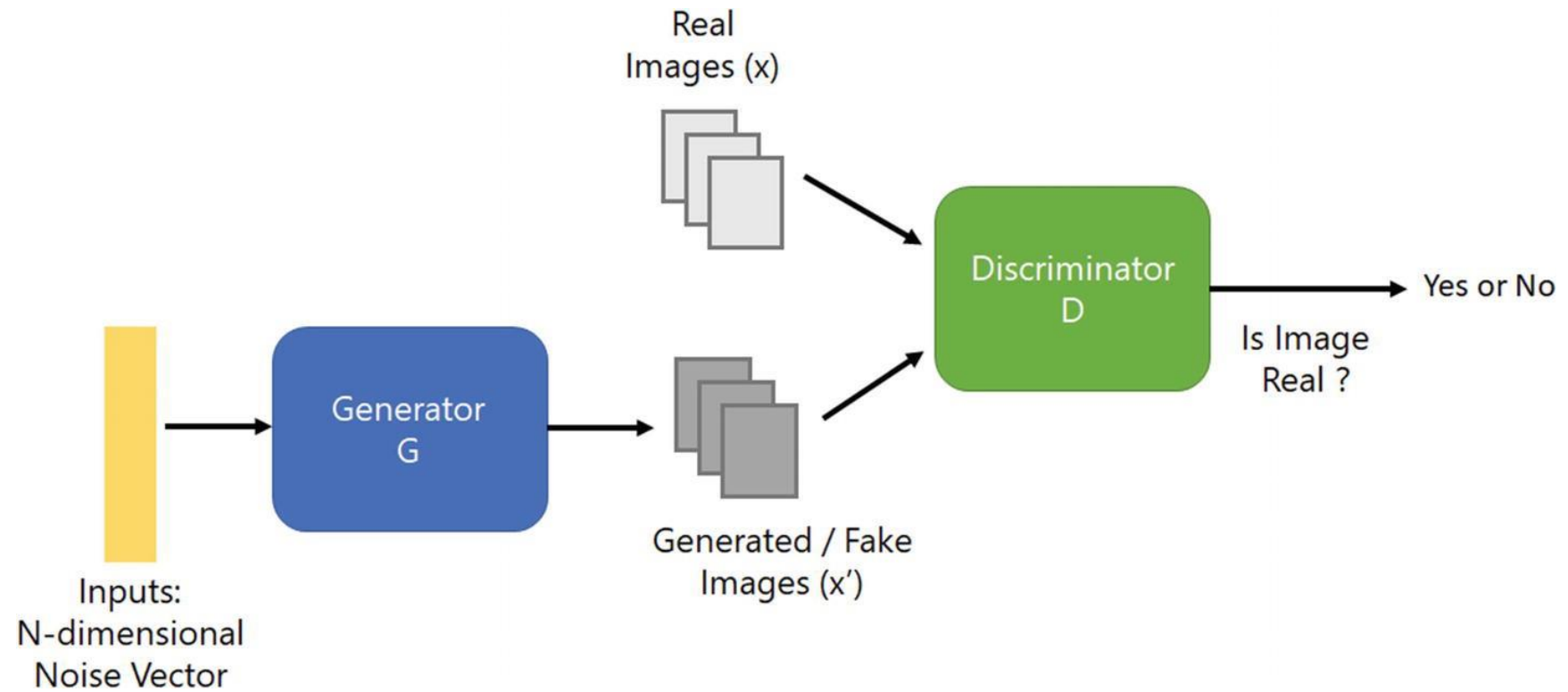
Reminder Contractive Denoising Autoencoders

- ▶ The true signal is always situated on a manifold inside the R^D space.
- ▶ Denoising autoencoder is trained to map a corrupted data point $\tilde{x}$ back to the original data point x .

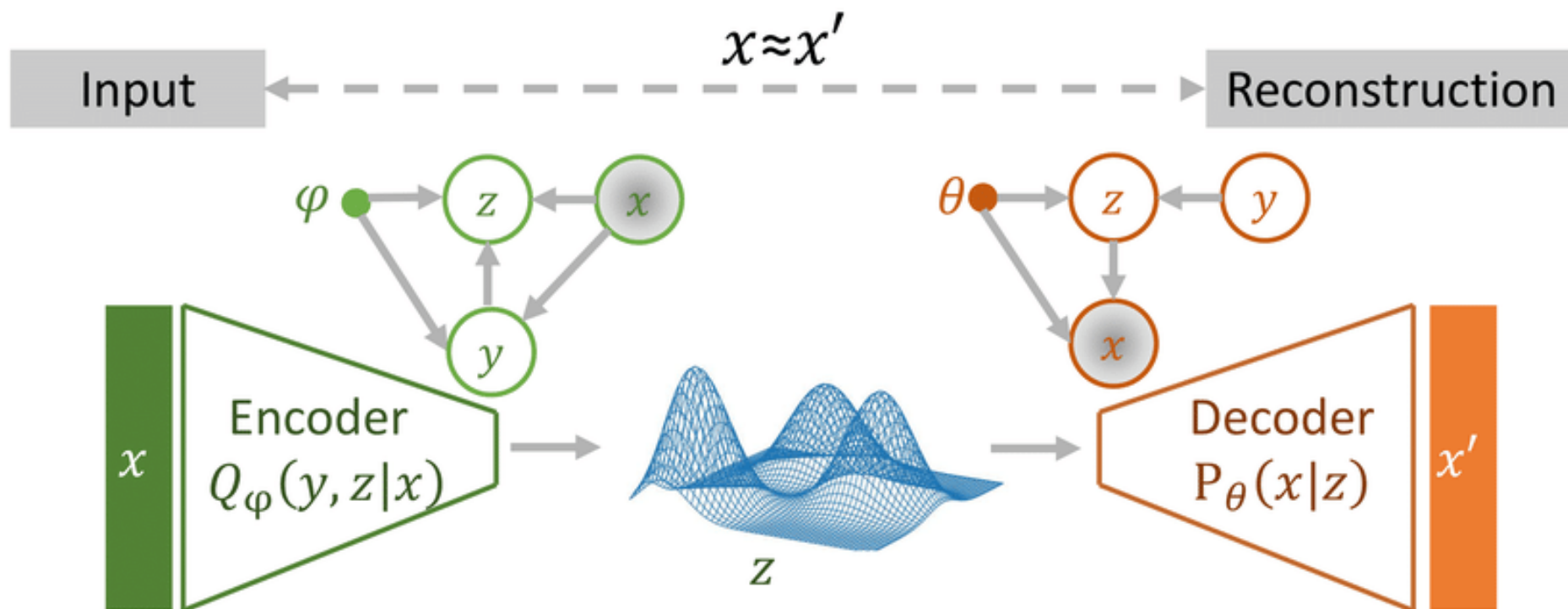


<https://arxiv.org/abs/1305.6663>

Vanilla GAN



VAE



$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{Q_\phi(y, z|x)} [\log P_\theta(x, y, z) - \log Q_\phi(y, z|x)]$$

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{Q_\phi(y, z|x)} \left[\underbrace{\log \frac{P(y)}{Q_\phi(y|x)}}_{\text{Entropy}} + \underbrace{\log \frac{P_\theta(z|y)}{Q_\phi(z|x, y)}}_{\text{Regularization}} + \underbrace{\log P_\theta(x|z)}_{\text{Reconstruction}} \right]$$

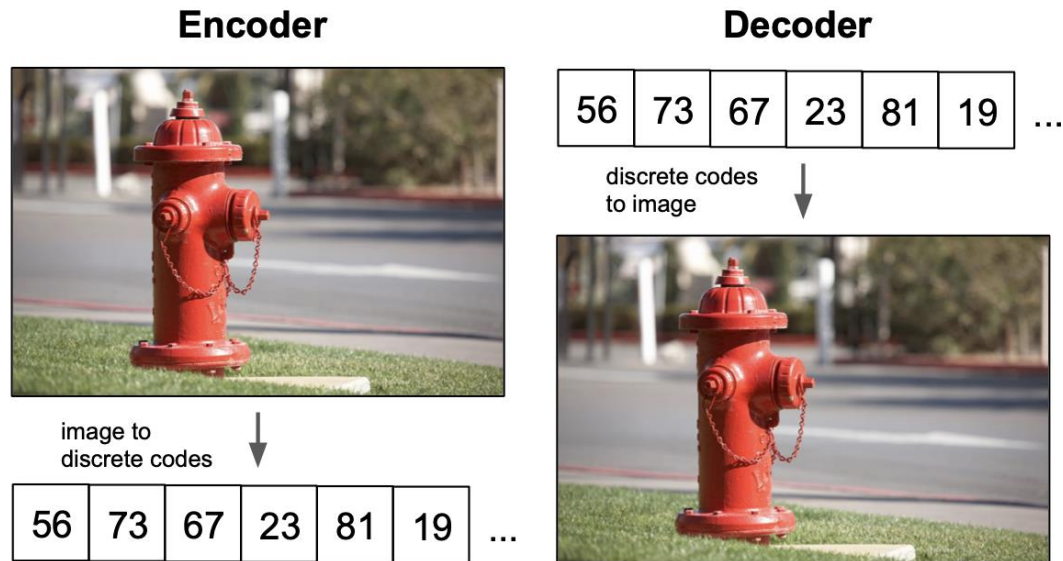
VQ-VAE



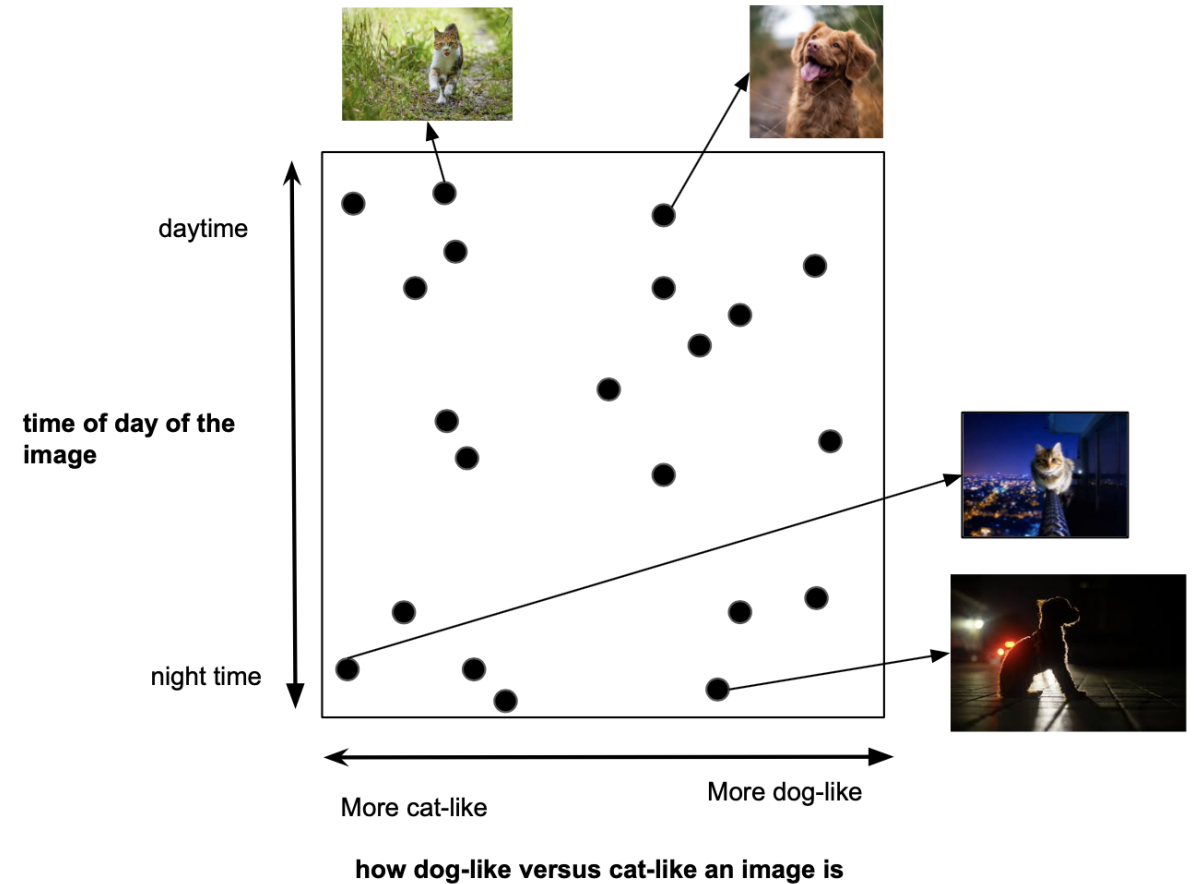
Latent Spaces

VAE gives a good representations

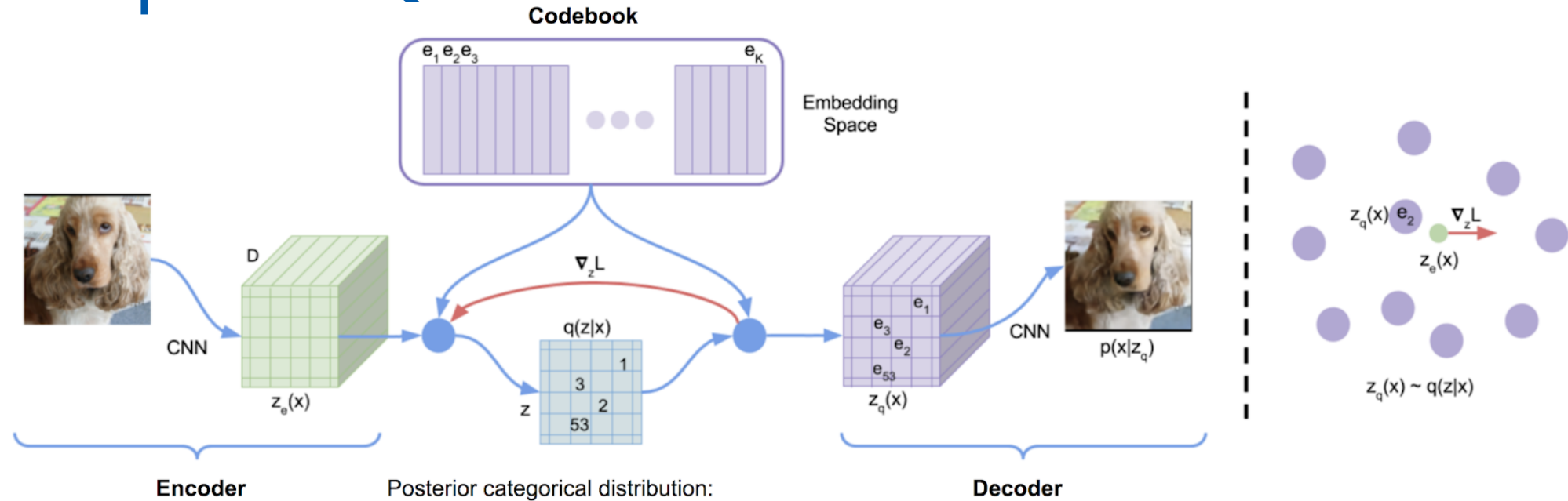
Can we study it?



An Oversimplified Example of a Cat/Dog Image Latent Space



Latent Space Quantization



- Reconstruction loss:

$$z_q(x) = \operatorname{argmin}_i \|z_e(x) - e_i\|$$

$z_e(x)$ - encoder vector for some raw input x

e_i - i -th codebook vector

$z_q(x)$ - resulting quantized vector that is passed as input to the decoder

VQ-VAE results

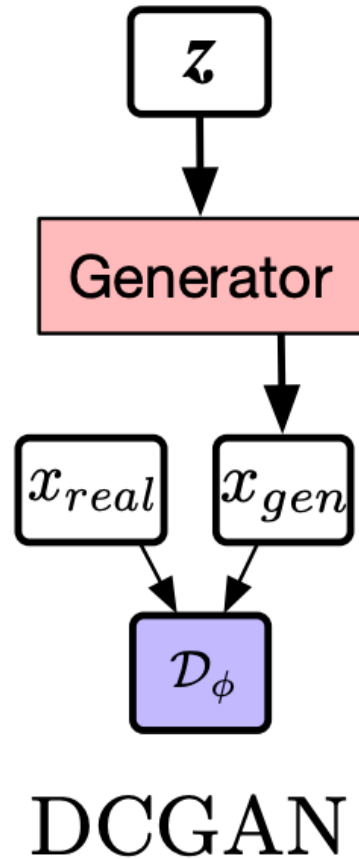


Figure 2: Left: ImageNet 128x128x3 images, right: reconstructions from a VQ-VAE with a 32x32x1 latent space, with $K=512$.

VAE+GAN

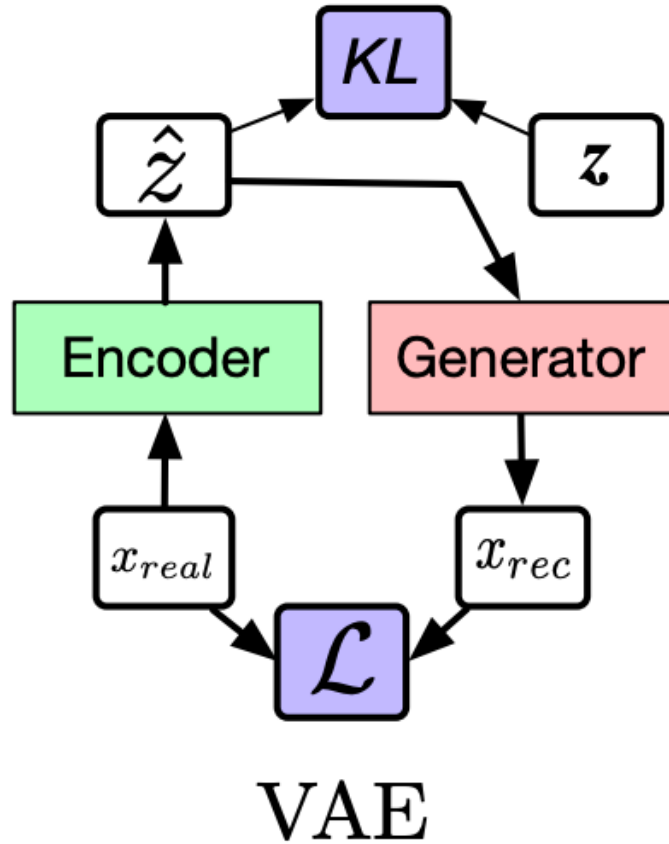


Improving GANs



- ▶ Main problems with GAN:
 - mode collapse;
 - intractable likelihood.
- ▶ Idea:
 - use likelihood-based mode;
 - have easier inference;
 - diversify sampling.

Reminder: VAE structure



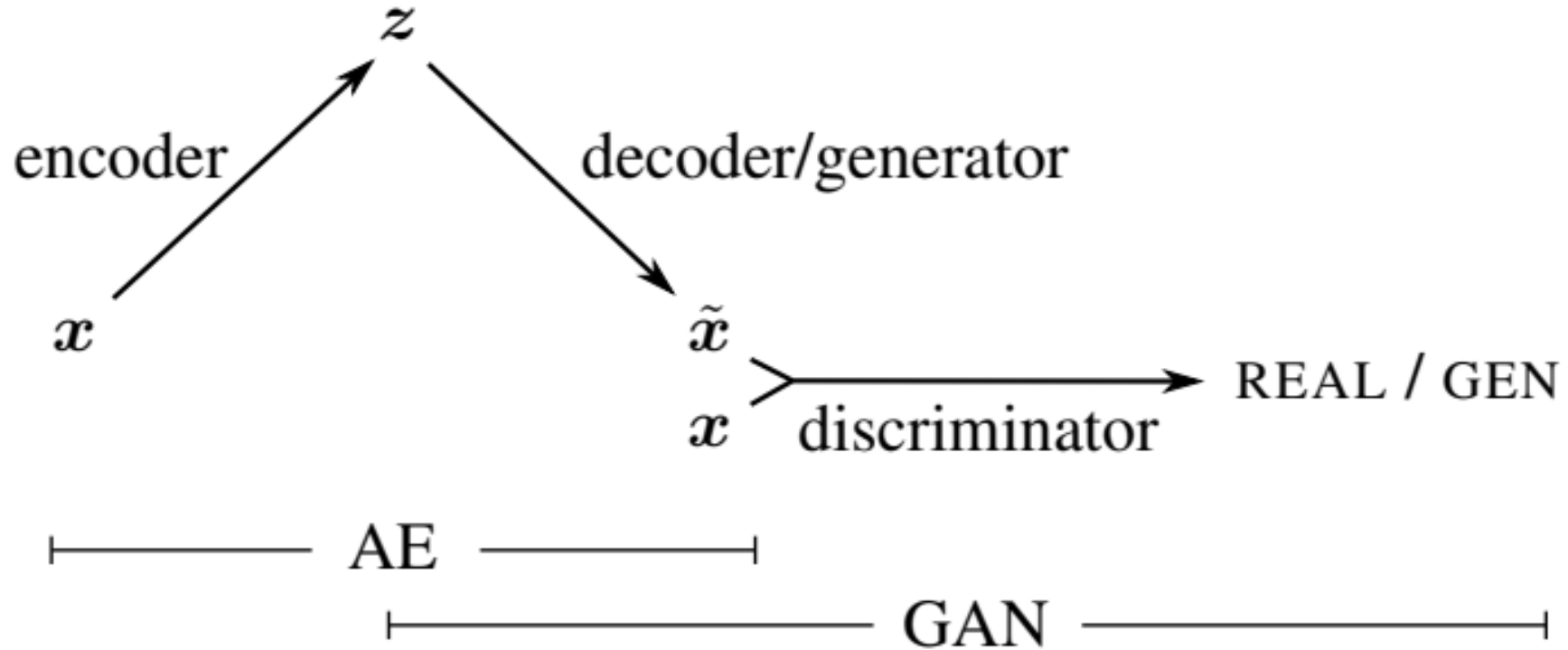
$$\mathcal{L}_{prior} = KL(q_n(z|x)||p(z))$$

+

$$-\mathcal{L}_{pixel} = \mathbb{E}_{q_n(z|x)}(\log p_{\theta}(x|z))$$

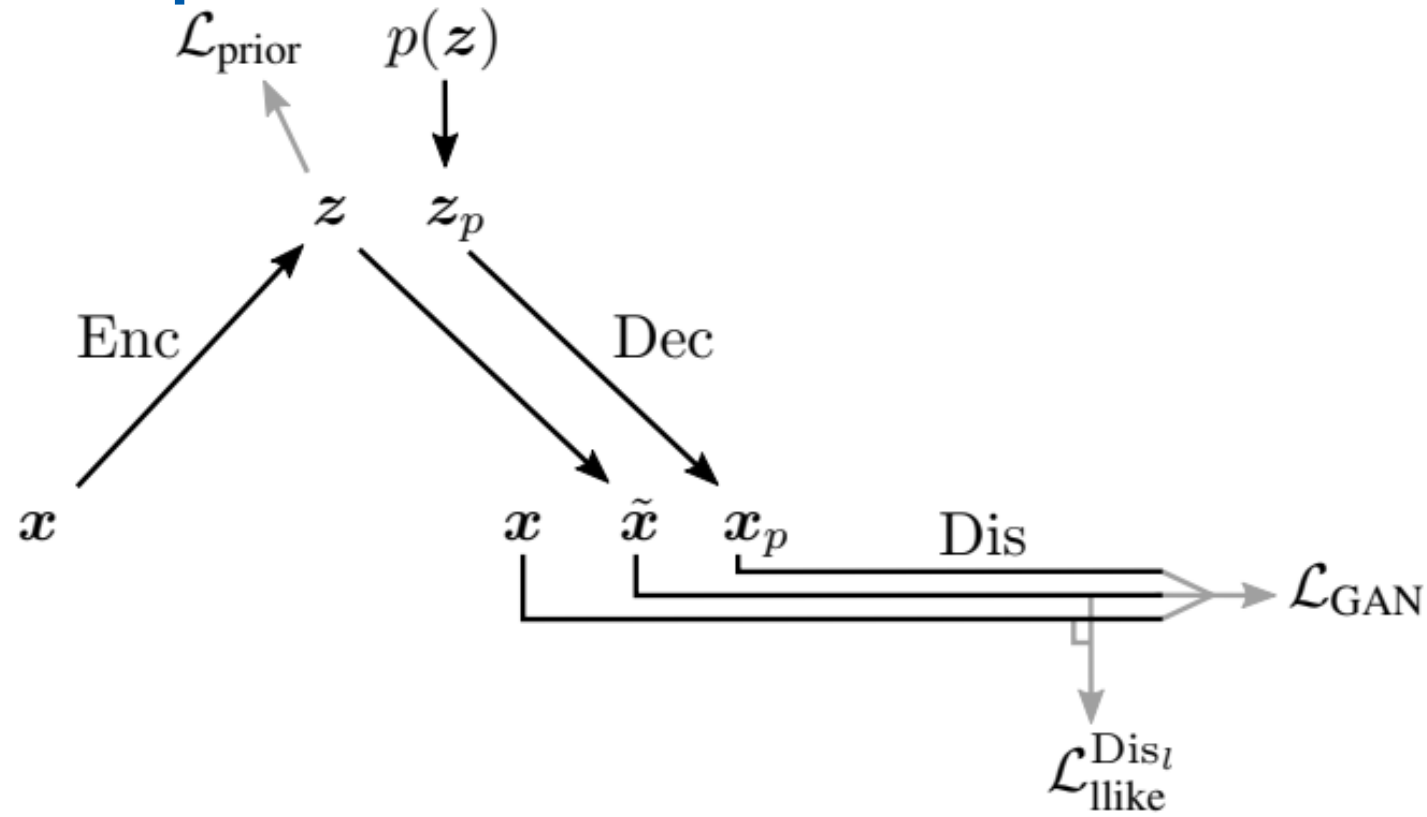
$$= \mathcal{L}_{VAE}$$

VAE+GAN



- ▶ problem of VAE is blurry output
- ▶ GAN overcomes due to the use of discriminator
- ▶ add GAN loss to VAE

VAE+GAN expanded

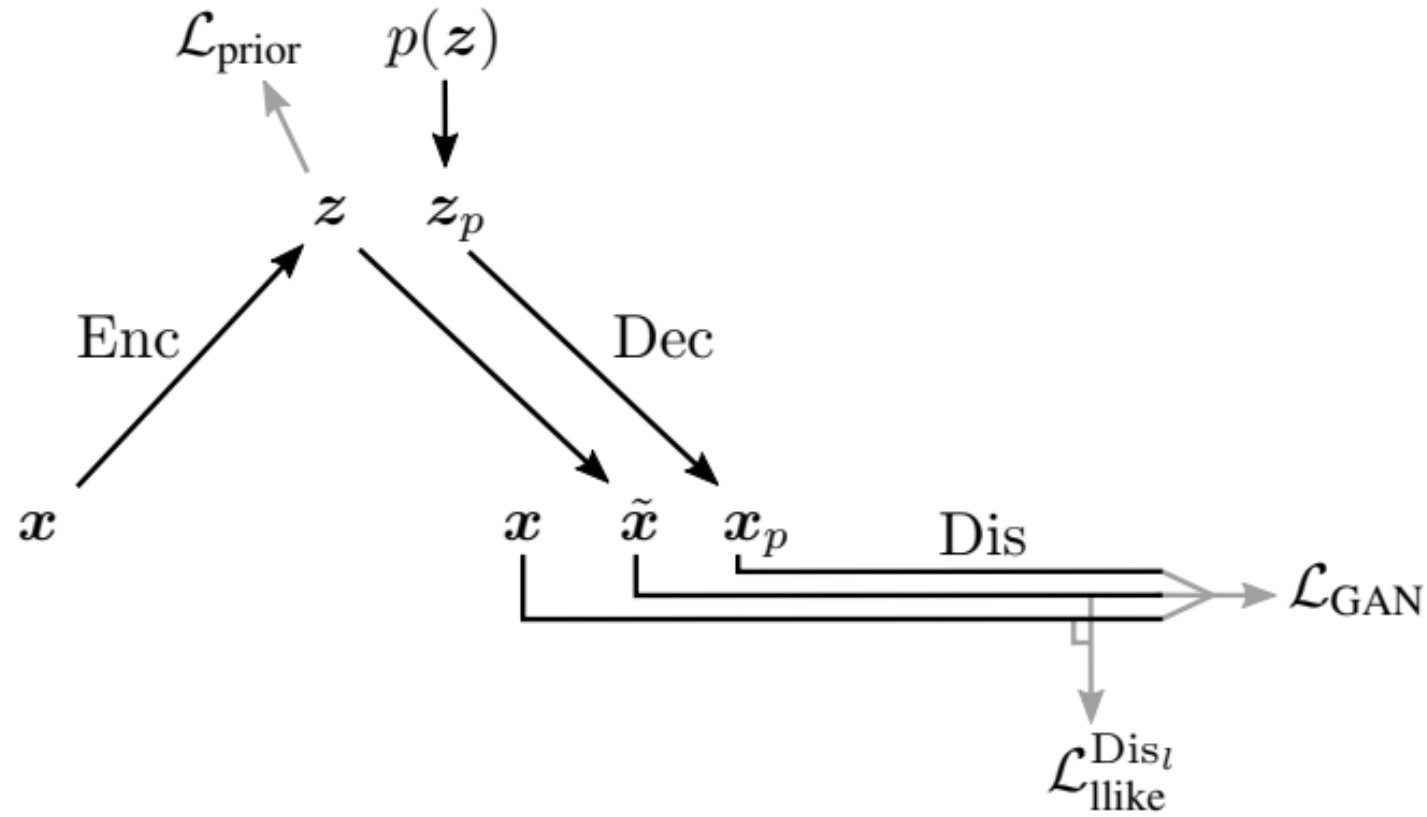


- Consider image as a whole:

$$-\mathcal{L}_{DIS} = \mathbb{E}_{q_n(z|x)} p(Dis_l(x)|z) = \mathbb{E}_{q_n(z|x)} N(Dis_l(x)|Dis_l(\tilde{x}), I), \quad \tilde{x} \sim Dec(z)$$

where $Dis_l(x)$ denote the hidden representation of the l -th layer of the discriminator.

VAE+GAN final loss



$$\mathcal{L}_{GAN+VAE} = \underbrace{\mathcal{L}_{prior} + \mathcal{L}_{DIS}}_{\text{Image}} + \underbrace{\mathcal{L}_{GAN}}_{\text{Style}}$$

VAE+GAN results

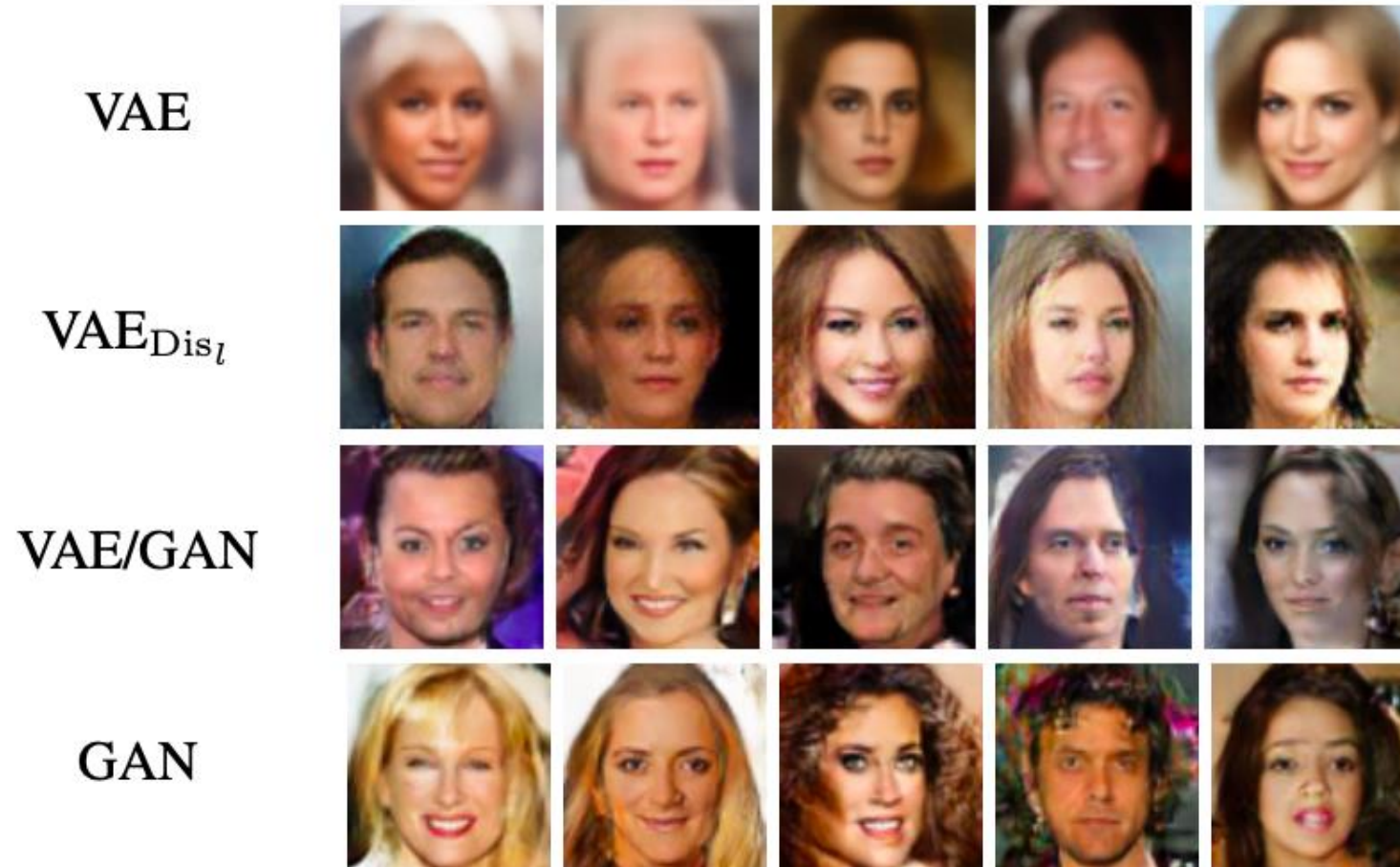


Figure 3. Samples from different generative models.

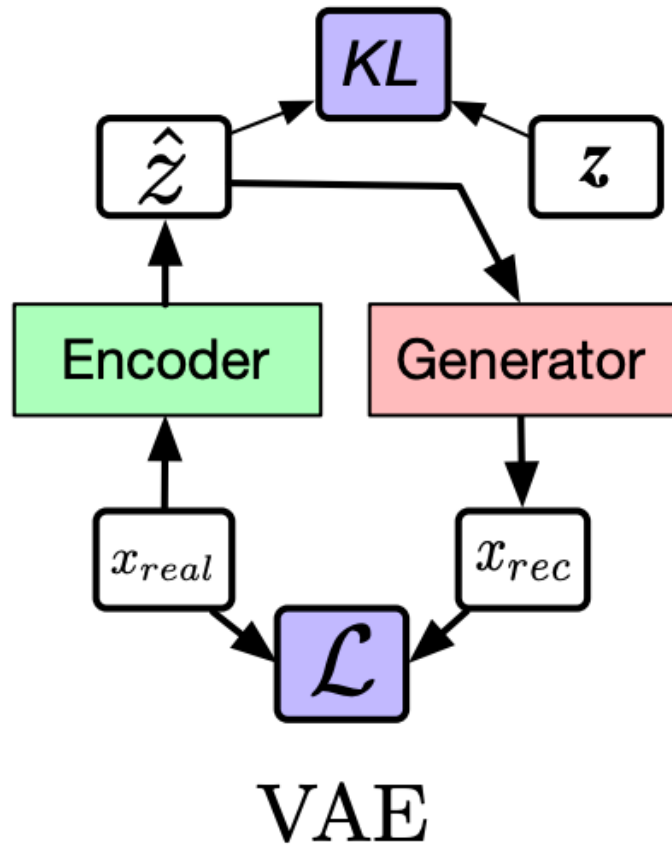
Wrap-up

- ▶ Benefits from combined loss and combined architecture.
- ▶ Better images than VAE or GAN standalone.
- ▶ Can be viewed as extension of VAE with adversarial selection.
- ▶ Can be viewed as extension of GAN with better latent space construction.

More VAEGANs



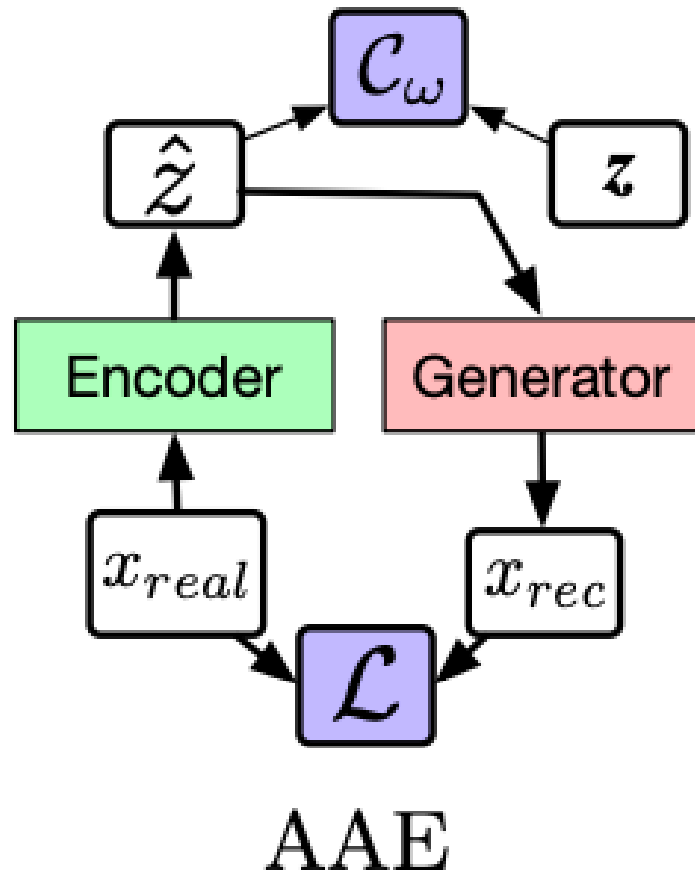
VAE structure



- ▶ KL divergence is considered as one of blurriness causes.
- ▶ We can switch this objective to GAN-like.

<https://arxiv.org/pdf/1511.05644.pdf>

AAE structure



- ▶ KL divergence is considered as one of blurriness causes.
- ▶ We can switch this objective to GAN-like.
- ▶ Price to pay: ELBO becomes less constrained.

<https://arxiv.org/pdf/1511.05644.pdf>

AAE results



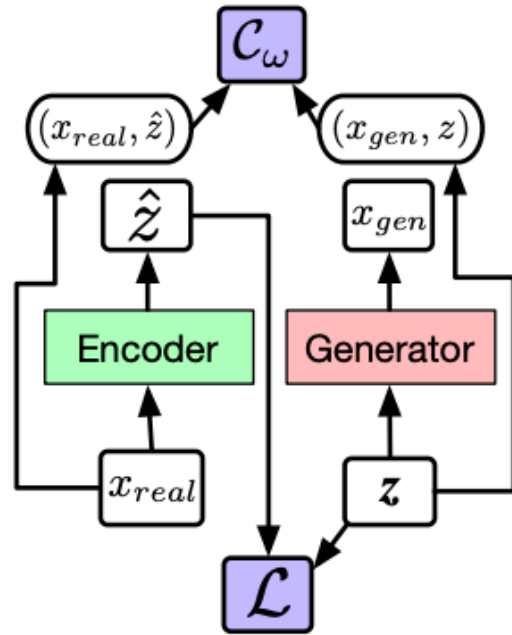
(b) VAE samples



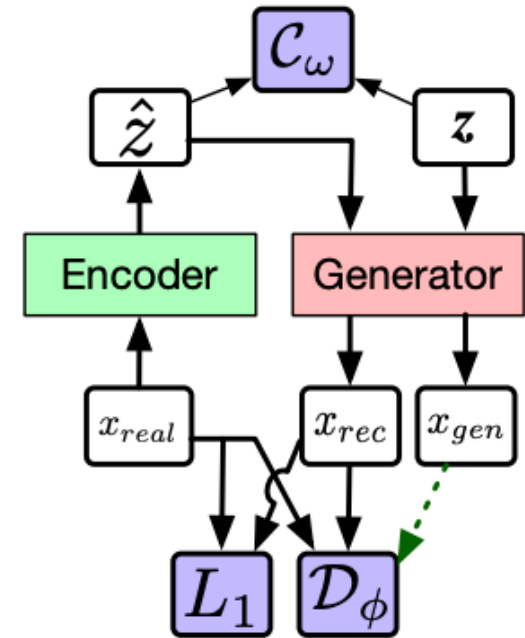
(c) AAE samples

<https://arxiv.org/pdf/1802.06847.pdf>

VEEGAN and VGH



VEEGAN



VGH/VGH++

- ▶ VGH: Marginal matching and implicit distributions using GANs both in latent and visible space;
- ▶ VEEGAN: Directly match in joint space.

VAE+GAN merged

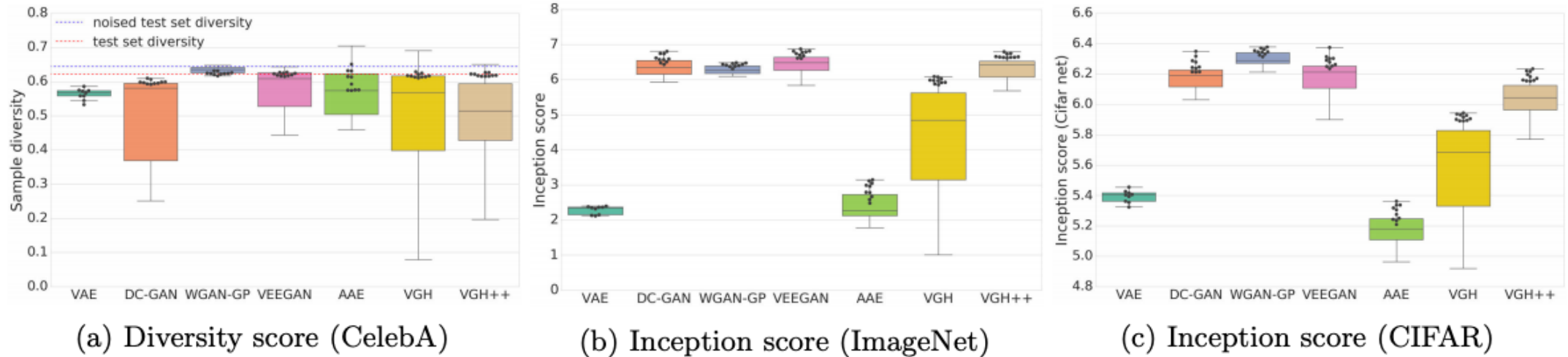


Figure 8: (Left) Sample diversity on CelebA, and is viewed relative to test set: too much diversity shows failure to capture the data distribution, too little is indicative of mode collapse. We also report the diversity obtained on a noised-version of the test set, which has a higher diversity than the test set. (Middle) Inception scores on CIFAR-10. (Right) Inception scores computed using a VGG-style network on CIFAR-10. For inception scores, higher values are better. For test data, diversity score: 0.621, inception score: 11.25, inception score (using CIFAR-10 trained net): 9.18. Best results are shown with black dots, and box plots show the hyperparameter sensitivity.

- ▶ Our aim was to improve GAN mode collapse and VAE blurriness.
- ▶ The results are yet to be improved.

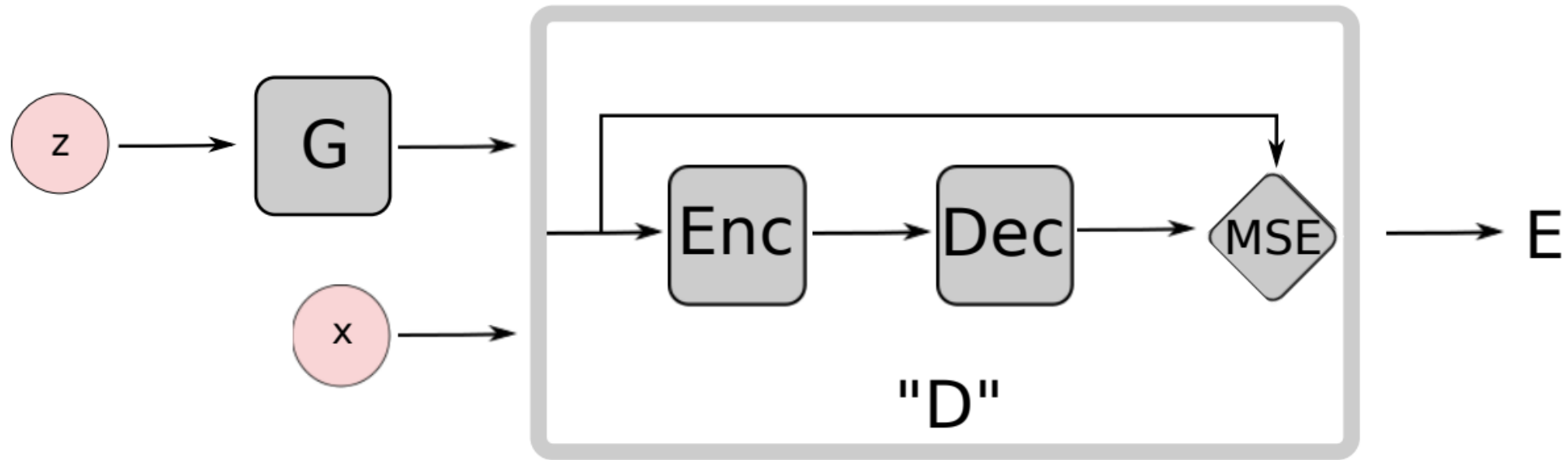
Conclusions

- ▶ VAE is one of the strong generative models that is used in current days.
- ▶ Several attempts were made to merge strong points of GANs and VAE.
- ▶ Work is ongoing.

Energy-based GAN



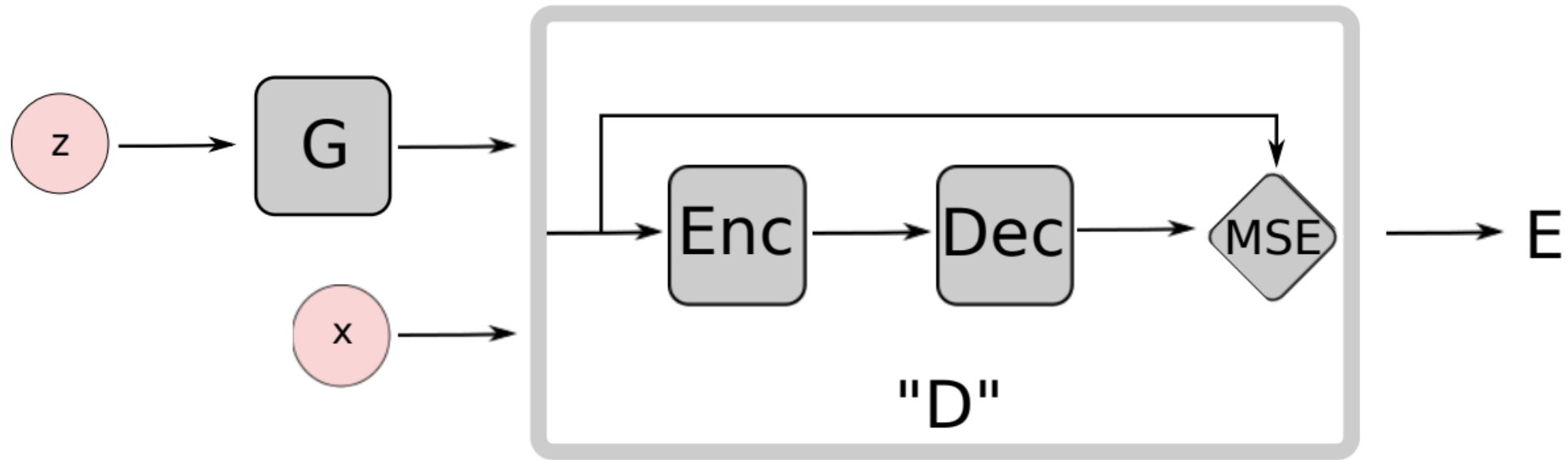
Autoencoder Discriminator



- Pros:
 - Takes into account the minibatch information properly.
 - Uses the full information about the manifold.

Junbo Zhao et al., ICLR 2017

Autoencoder Discriminator



- Use AE to extract latent features of the input image by an encoder and reconstruct it again with the decoder with MSE loss:

$$D(x) = ||Dec(Enc(x)) - x||$$

EB-GAN training

For $[\cdot]^+ = \max(0, \cdot)$:

$$\mathcal{L}_D(x, z) = D(x) + [m - D(G(z))]^+;$$

$$\mathcal{L}_G(x, z) = D(G(z)),$$

- ▶ **autoencoder**: reconstruction cost $D(x)$ for real images is low;
- ▶ $D(x)$ is trained first several rounds;
- ▶ once $G(z)$ generates sufficiently good images $D(x)$ training resumes;
- ▶ repelling loss to address AE collapse problem:

$$f_{PT} = \frac{1}{N(N-1)} \sum_i \sum_{j \neq i} \frac{S_i^T S_j}{\|S_i\| \|S_j\|}.$$

where $S \in \mathbb{R}^{s \times N}$ a batch of sample of size N representations taken from the encoder output layer.

GAN energy interpretation

For $[\cdot]^+ = \max(0, \cdot)$:

$$\mathcal{L}_D(x, z) = D(x) + [m - D(G(z))]^+;$$

$$\mathcal{L}_G(x, z) = D(G(z)),$$

- ▶ **autoencoder**: reconstruction cost $D(x)$ for real images is low;
- ▶ $D(x)$ does not have probability interpretation;
- ▶ one can use **energy interpretation** instead.

EB-GAN results



Figure 4: Generation from the grid search on MNIST. Left(a): Best GAN model; Middle(b): Best EBGAN model. Right(c): Best EBGAN-PT model.

Boundary equilibrium GAN



Wasserstein Distance lower bound

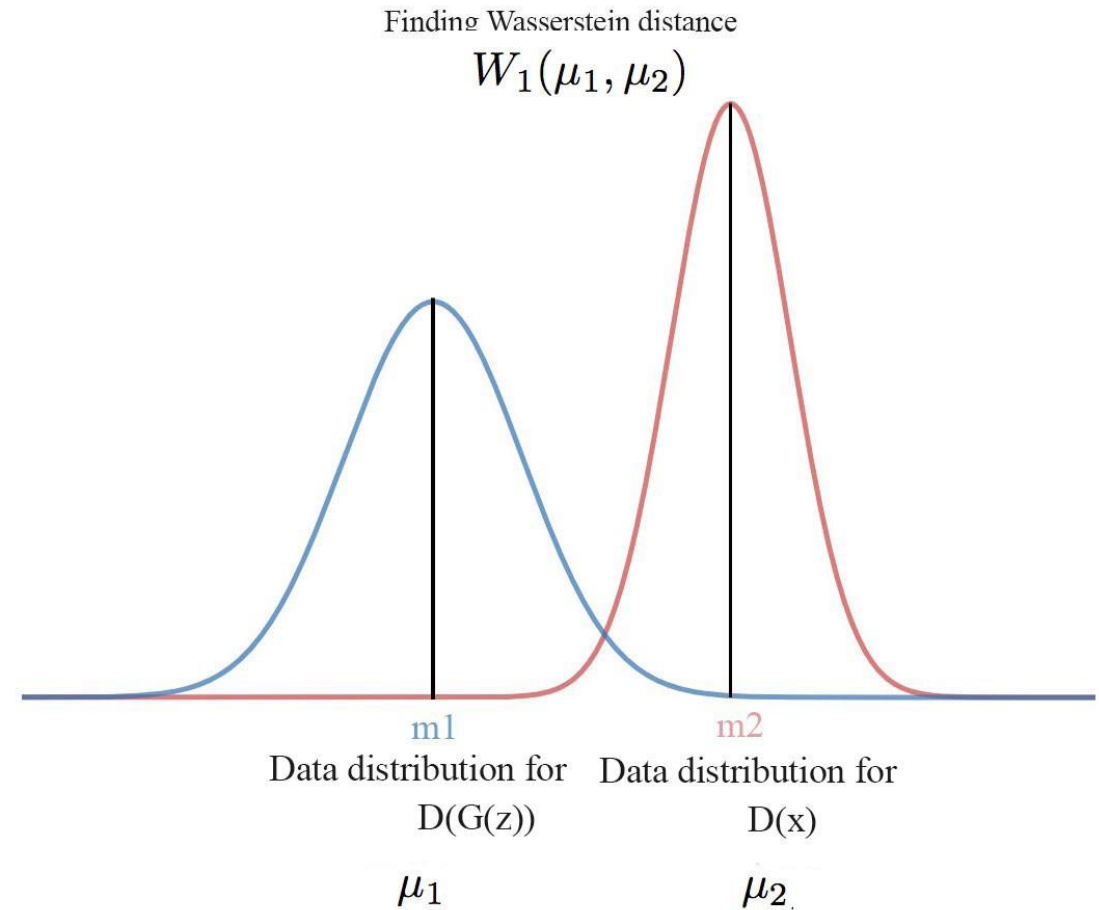
- **Wasserstein distance:**

$$W(\mu_1, \mu_2) = \inf_{\gamma \in \Pi} \mathbb{E}_{(x,y) \sim \gamma} \|x - y\|$$

- **Jensen's inequality:**

$$\begin{aligned} W(\mu_1, \mu_2) &\geq \inf_{\gamma \in \Pi} |\mathbb{E}_{(x,y) \sim \gamma} x - y| = \\ &= |m_1 - m_2|, \end{aligned}$$

where m_i are the mean on μ_i .



Wasserstein Discriminator

- ▶ We have $D(x)$ as AE:

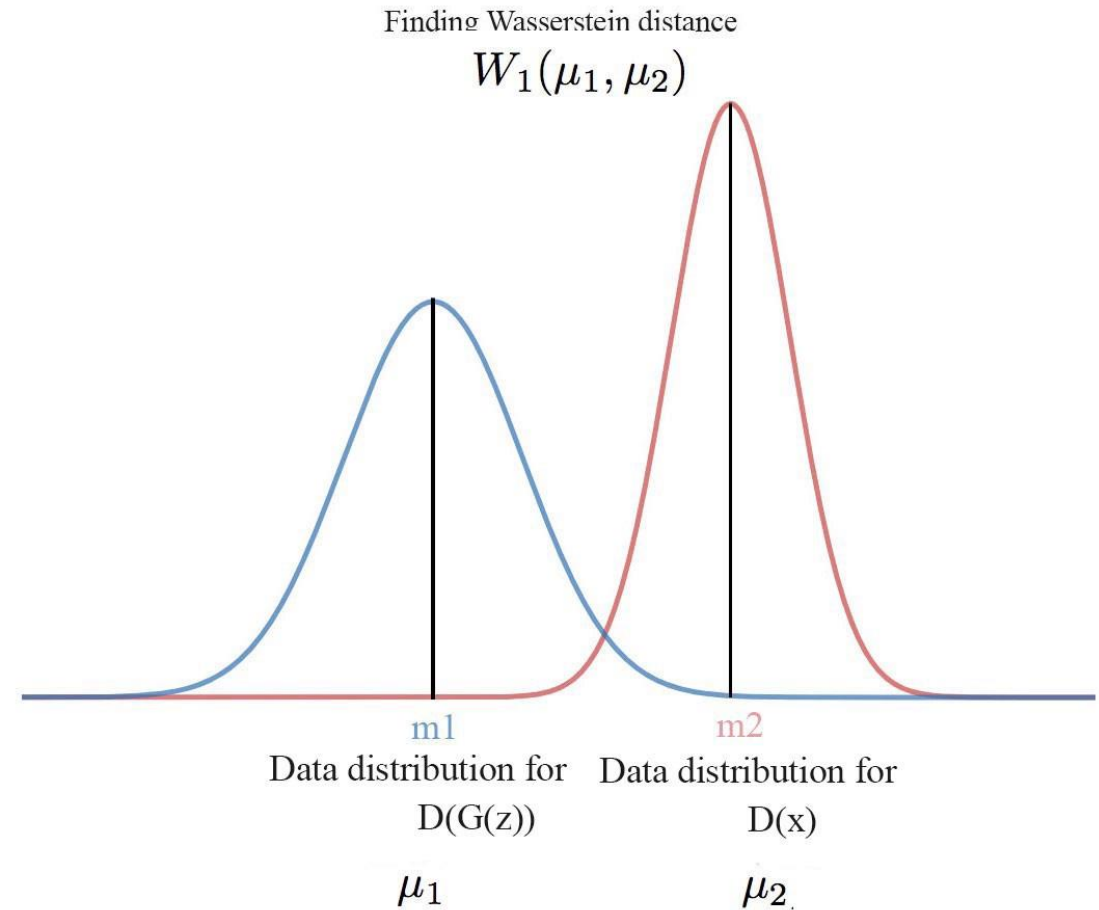
$$D(x) = ||Dec(Enc(x)) - x||$$

$$\mathcal{L}_D = W(\mu_1, \mu_2) \geq |m_1 - m_2|.$$

- ▶ We can use $D(x)$ in minibatch instead of mean:

$$\mathcal{L}_D = D(x) - D(G(z)).$$

- ▶ We thus optimize W between losses.
- ▶ No need for K-Lipshitz, since no Kantorovich-Rubinstein duality is used.



Equilibrium term

- ▶ we need to maintain balance between G and D :

$$\mathbb{E}(D(x)) = \mathbb{E}(D(G(z)))$$

- ▶ we thus can use a parameter to balance the impact:

$$\gamma = \frac{\mathbb{E}(D(x))}{\mathbb{E}(D(G(z)))}.$$

- ▶ γ can be chosen to sharpen the image.

BEGAN formulation

- ▶ We thus can write full optimization for BEGAN

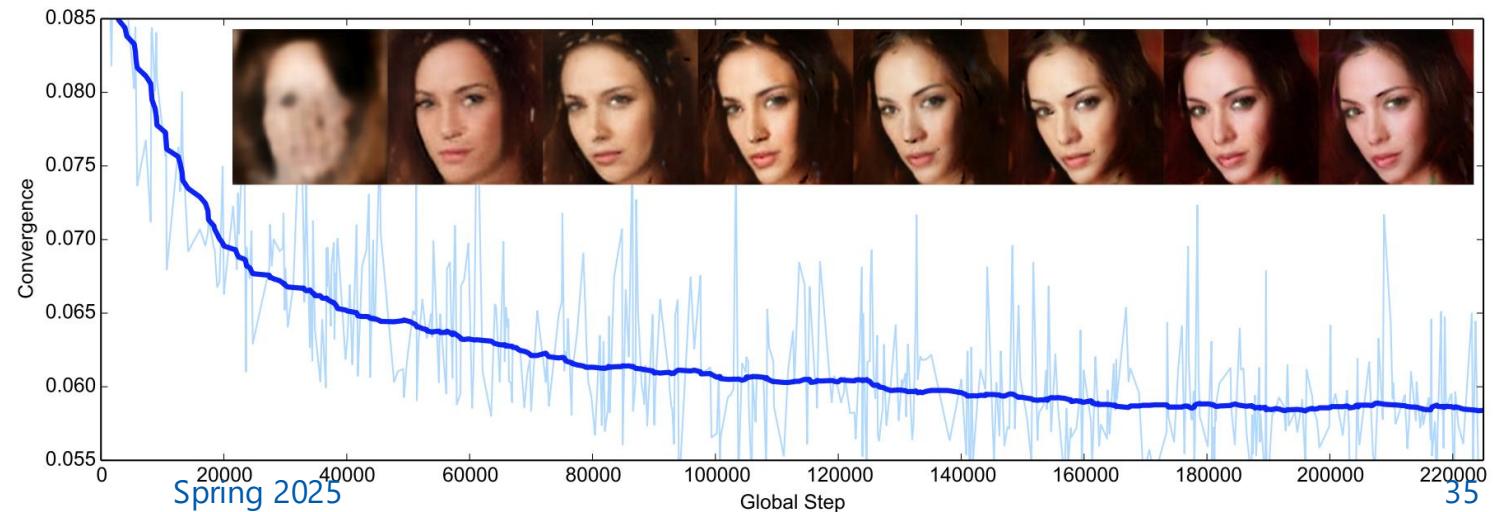
$$\mathcal{L}_D = D(x) - k_t D(G(z));$$

$$\mathcal{L}_G = D(G(z));$$

$$k_{t+1} = k_t + \lambda_k (\gamma D(x) - D(G(z))).$$

- ▶ Dropping γ leads to mode collapse.
- ▶ To monitor the convergence:

$$M_{global} = D(x) + (\gamma D(x) - D(G(z)))$$



BEGAN results



(c) Our results (128x128 with 128 filters)



(d) Mirror interpolations (our results 128x128 with 128 filters)

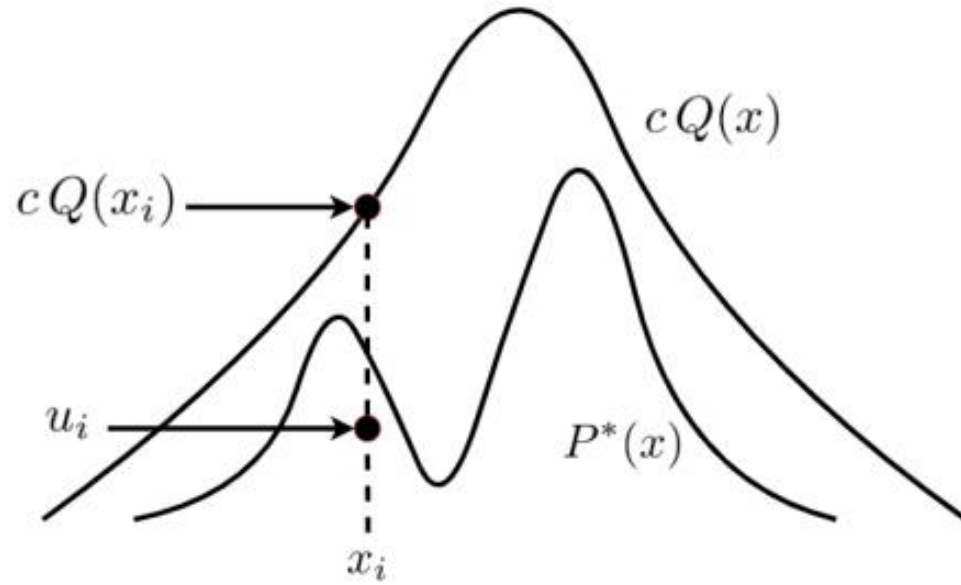
Wrap-up

- ▶ We can change the architecture of discriminator.
- ▶ This might lead to change of the optimization idea.
- ▶ If we use autoencoder as discriminator we have access to the energy instead of probability.
- ▶ We can optimize Wasserstein distance not only for datasets, but also for results of function.

Rejection Sampling



Rejection Sampling



```
1 Input:  $P^*(X), Q(X), c$   
2 Output:  $\mathcal{S} = \{x_i\}_{i=1}^n \sim P^*(X)$   
3  $\mathcal{S} \leftarrow \emptyset$   
4 for sample index  $i$  from 1 to  $n$  do  
5    $x_i \sim Q(X)$   
6    $u_i \sim U(0, cQ(x_i))$   
7   if  $u_i < P^*(x_i)$  then  
8     | Accept  $x_i$ :  $\mathcal{S} \leftarrow \mathcal{S} \cup \{x_i\}$   
9   else  
10  | Reject  $x_i$ :  $i \leftarrow i - 1$ 
```

Ideal Discriminator

- ▶ Ideal discriminator:

$$D^*(x) = \frac{p(x)}{p(x) + q_\theta(x)}.$$

- ▶ Remember f-GAN idea of last layer:

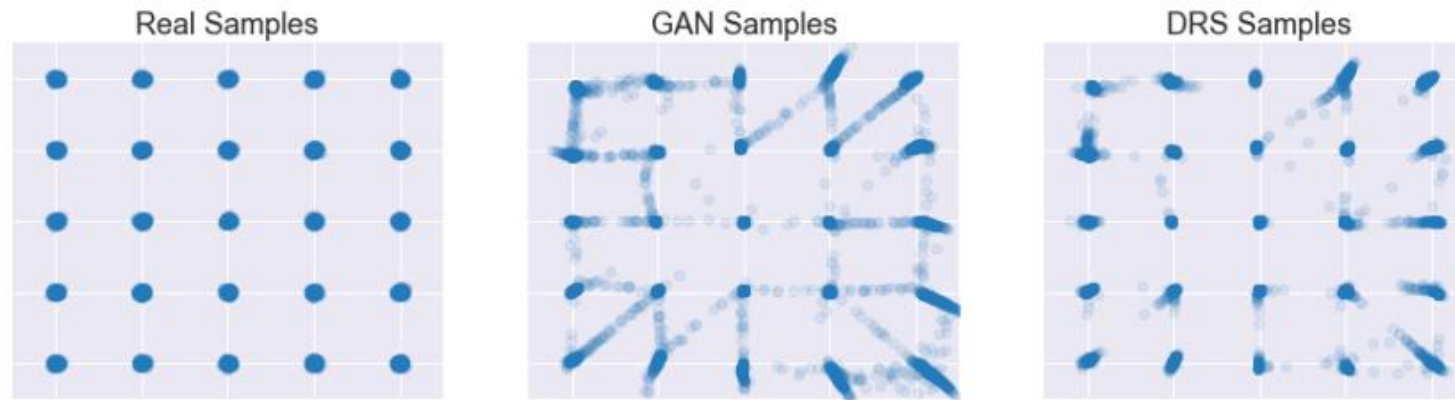
$$D^*(x) = \frac{1}{1 + e^{-\tilde{D}^*(x)}} = \frac{p(x)}{p(x) + q_\theta(x)}.$$

- ▶ Thus:

$$\frac{p(x)}{q_\theta(x)} = e^{\tilde{D}^*(x)}.$$

- ▶ This defines constant for rejection sampling.

Discriminator Rejection Sampling



ImageNet	IS	FID
Without DRS	52.34 ± 0.45	18.21 ± 0.14
With DRS	61.44 ± 0.09	17.14 ± 0.09

Results suggest that the quality of sampling is improved

Your GAN is secretly an energy based model

- ▶ Previous results can be revisited (with acceptance probability):

$$\frac{p(x)}{q_{\theta}(x)} = e^{\tilde{D}^*(x)}.$$

- ▶ And applied to the latent space. This will create a rule for new latent space distribution:

$$p_t(z) = p_0(z)r(z)/C.$$

- ▶ Which can be rewritten as:

$$p_t(z) = e^{-E(z)} / Z_0, \text{ with tractable } E(z):$$

$$E(z) = -\log p_0(z) - \tilde{D}(G(z)).$$

- ▶ This can be used to define MCMC in latent space and later obtain $x \sim G(z)$.

Energy-based sampling: results



Figure 4. Top-5 nearest neighbor images (right columns) of generated samples (left column).

Discussion

- ▶ GAN's discriminator can enable better modeling of the data distribution with Discriminator Driven Latent Sampling.
- ▶ The major advantage of DDLS is that it allows MCMC sampling in the latent space.

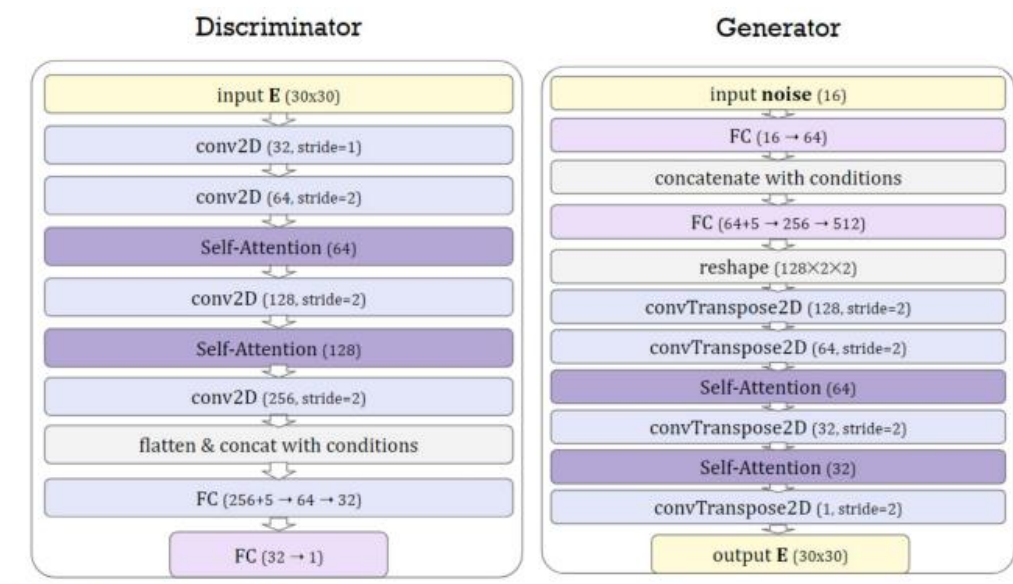
Implementing GANs



Motivation

- ▶ GANs, being the most popular generative models, often aim for production.
- ▶ The sampling speed thus becomes important.
- ▶ Easiest way: implement model description.

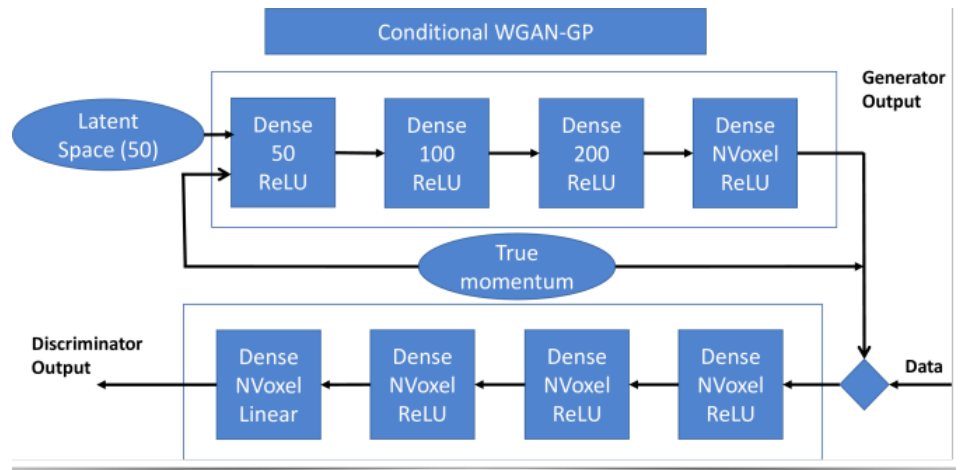
Rogachev et al., ACAT 2021
Ahmed et al., CERN seminar'2022



EPJ Web of Conferences 251, 03043 (2021)

VS

100X

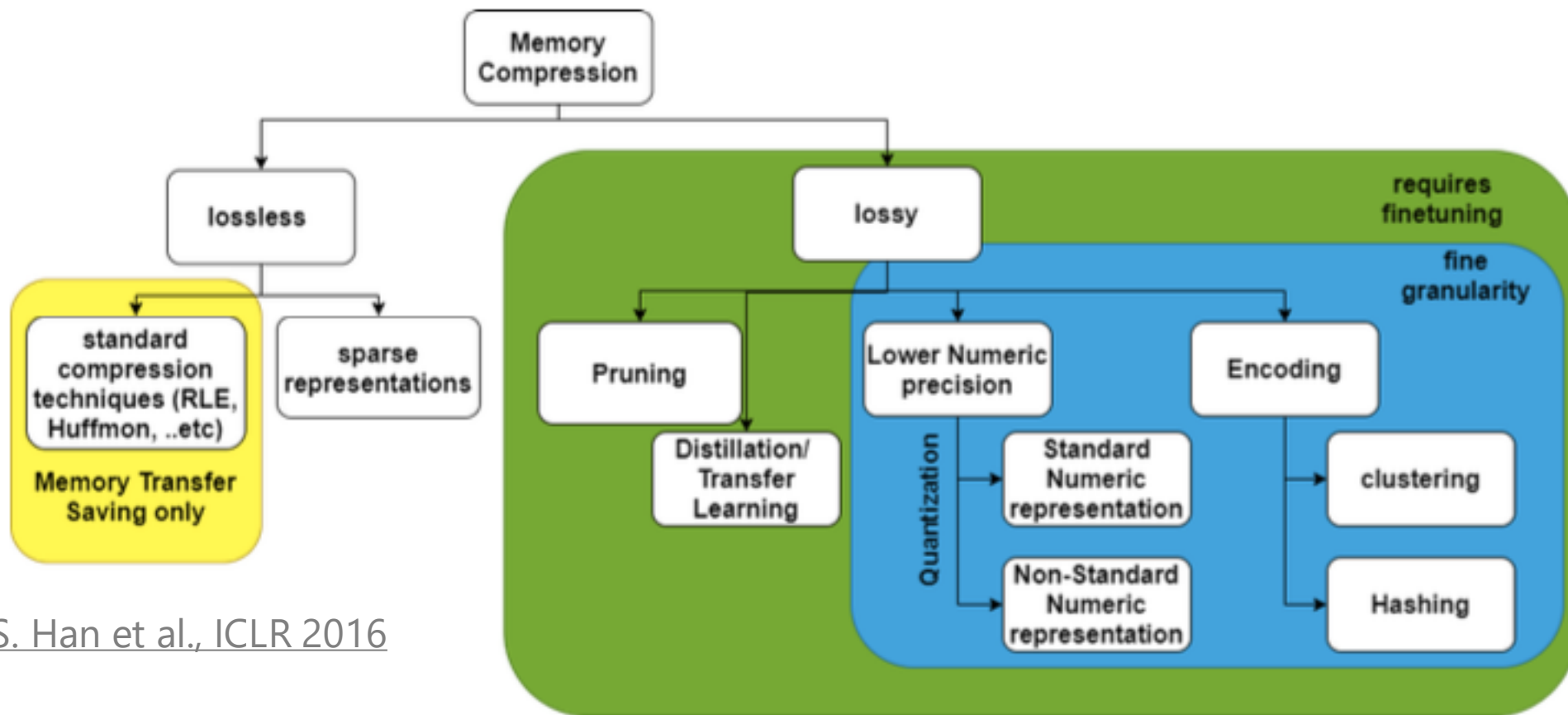


Acceleration techniques

- ▶ **Memory compression** – minimize memory requirements.
- ▶ **Computation optimization** – decrease number of mathematical operations.
- ▶ **Dataflow optimization** – maximize data reuse and minimize ineffectual operations.

D. Tantawy et al., A survey on GAN acceleration using memory compression techniques, Journal of Engineering and Applied Sciences, 2021

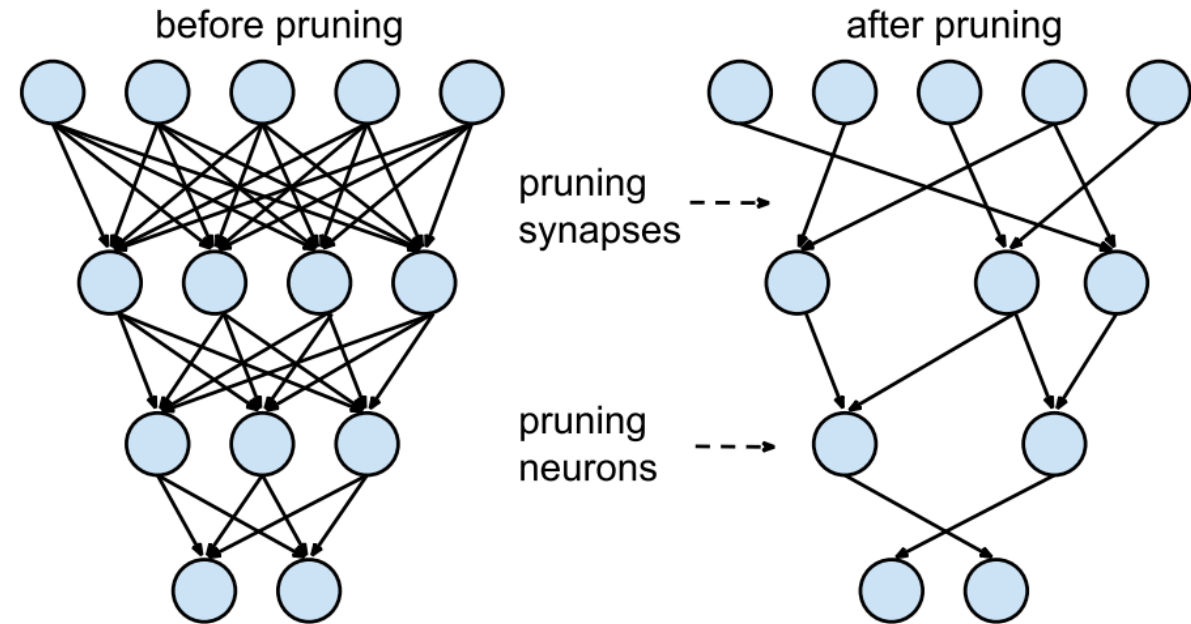
Memory Compression



S. Han et al., ICLR 2016

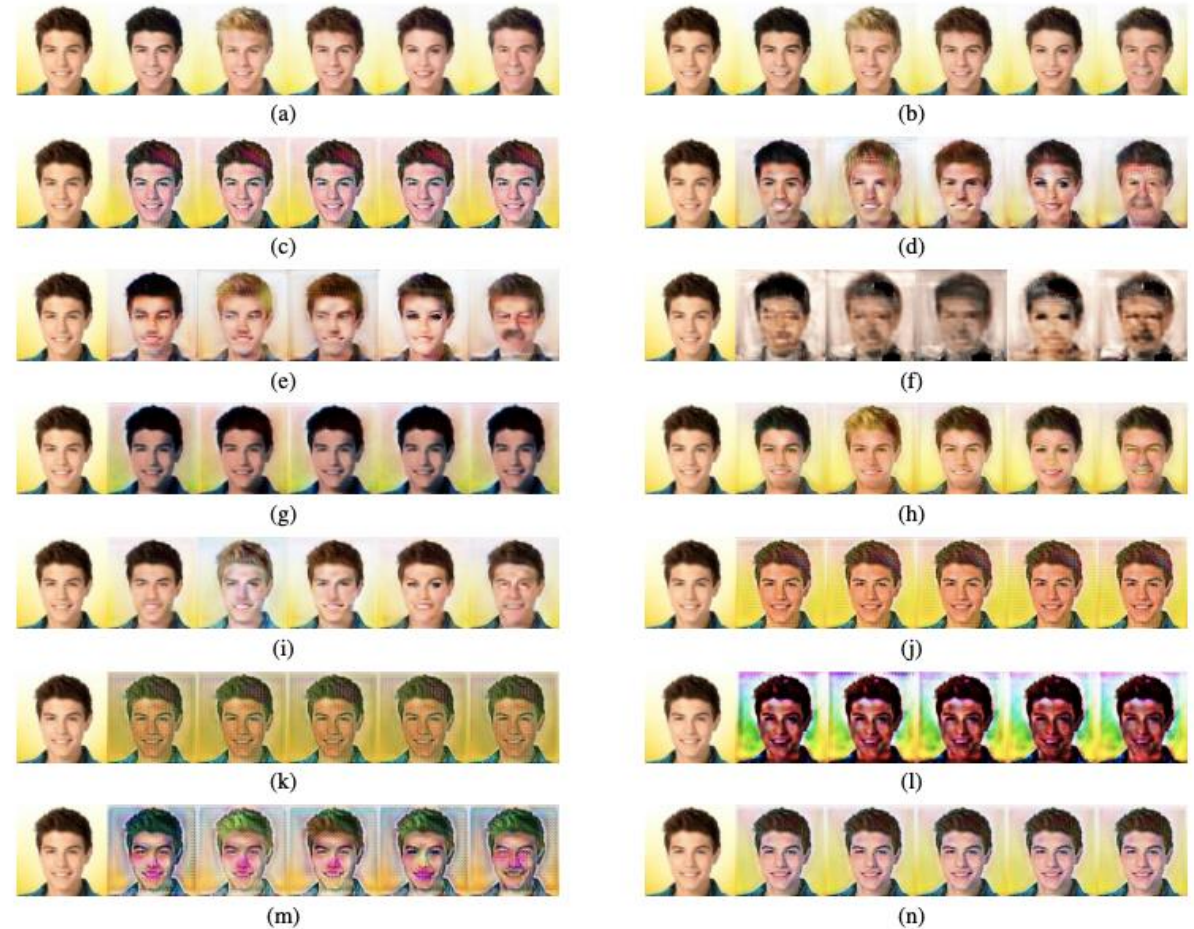
Pruning Decisions

- ▶ **Criteria**
 - random;
 - threshold based;
 - evolutionary.
- ▶ **Granularity**
 - structured;
 - unstructured.
- ▶ **Application**
 - Before/after training.
- ▶ **D vs. G**



Pruning for GANs

- Out-of-the-box techniques fail:
 - weak evaluation metrics;
 - unstable training;
 - high-dimensional input/output spaces.



C. Yu et al., Self-Supervised GAN compression, NeurIPS'20

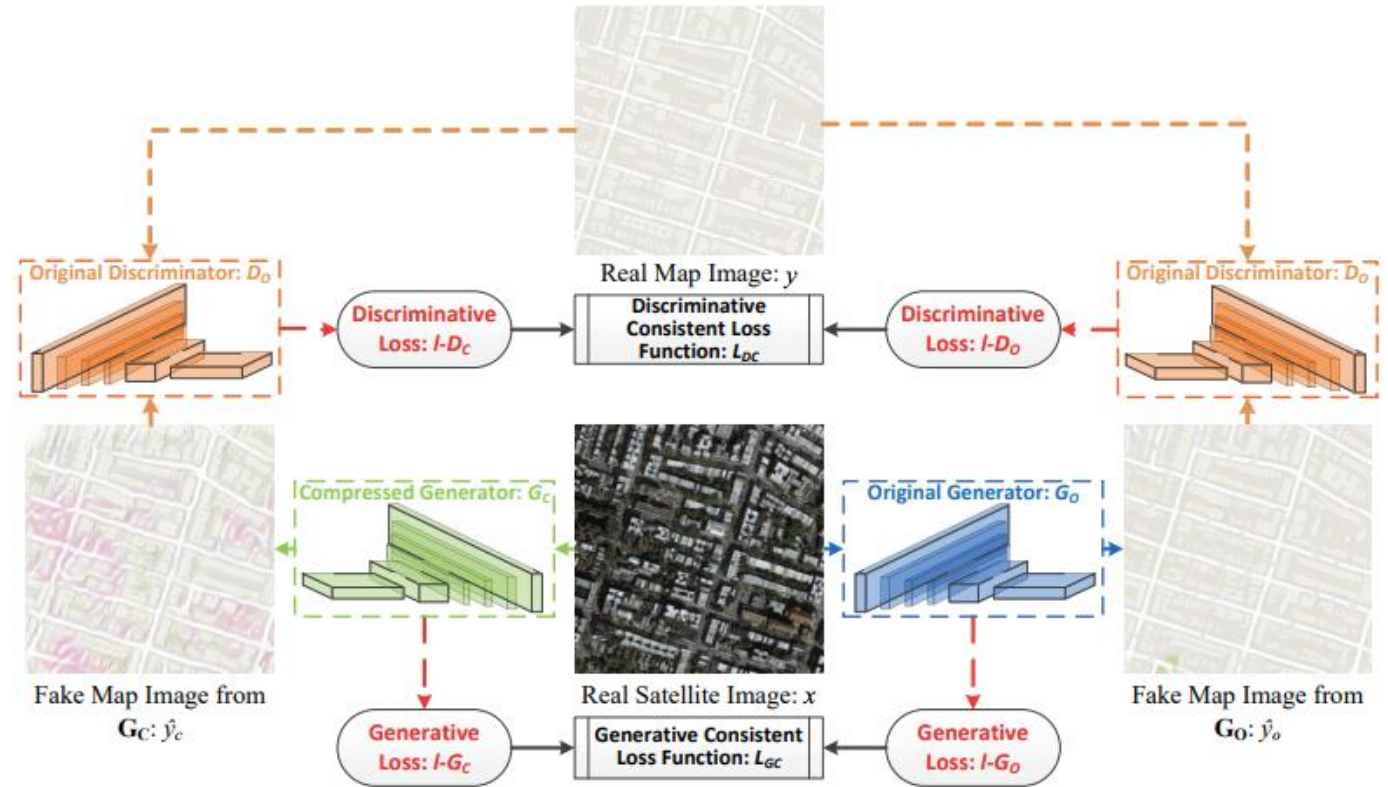
Self-supervised Pruning

- Use discriminator from the uncompressed GAN to train compressed one.

$$L_{GC}(l-G_O, l-G_C) = |l-Gen_O - l-Gen_C|/|l-Gen_O| + \alpha |l-Cla_O - l-Cla_C|/|l-Cla_O| + \beta |l-Rec_O - l-Rec_C|/|l-Rec_O|$$

$$L_{DC}(l-D_O, l-D_C) = |l-Dis_O - l-Dis_C|/|l-Dis_O| + \delta |l-GP_O - l-GP_C|/|l-GP_O|$$

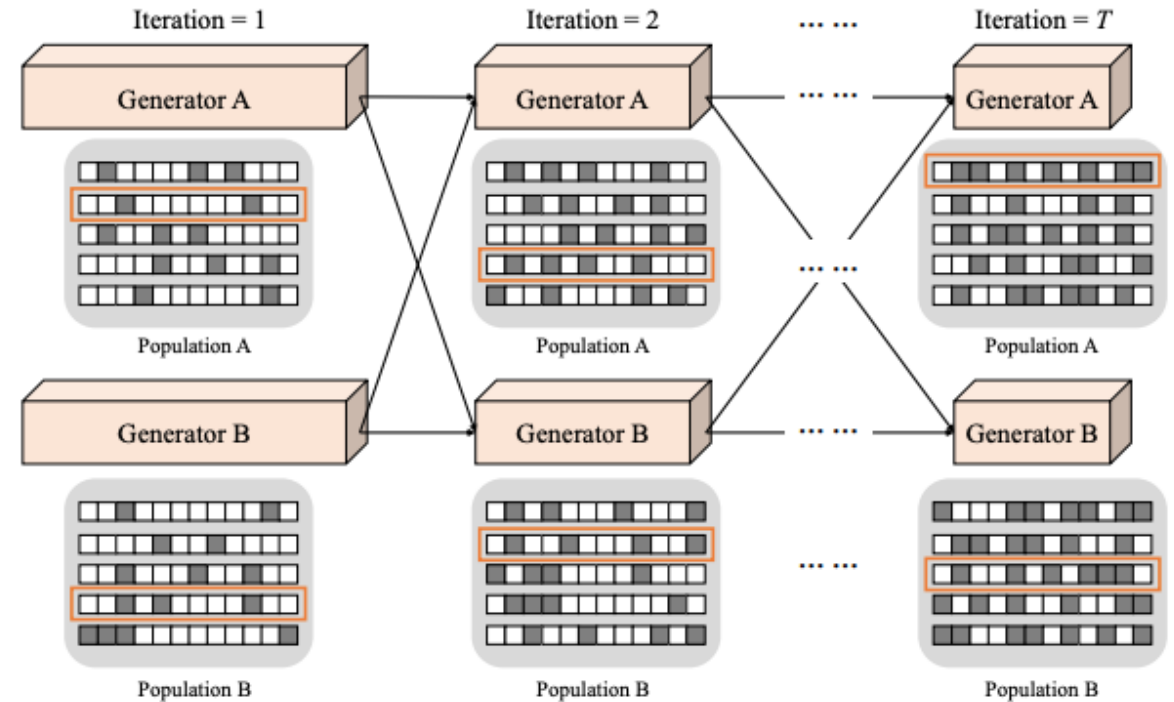
$$L_{Overall} = L_{GC}(l-G_O, l-G_C) + \lambda L_{DC}(l-D_O, l-D_C)$$



C. Yu et al., Self-Supervised GAN compression, NeurIPS'20

Co-evolutionary Pruning

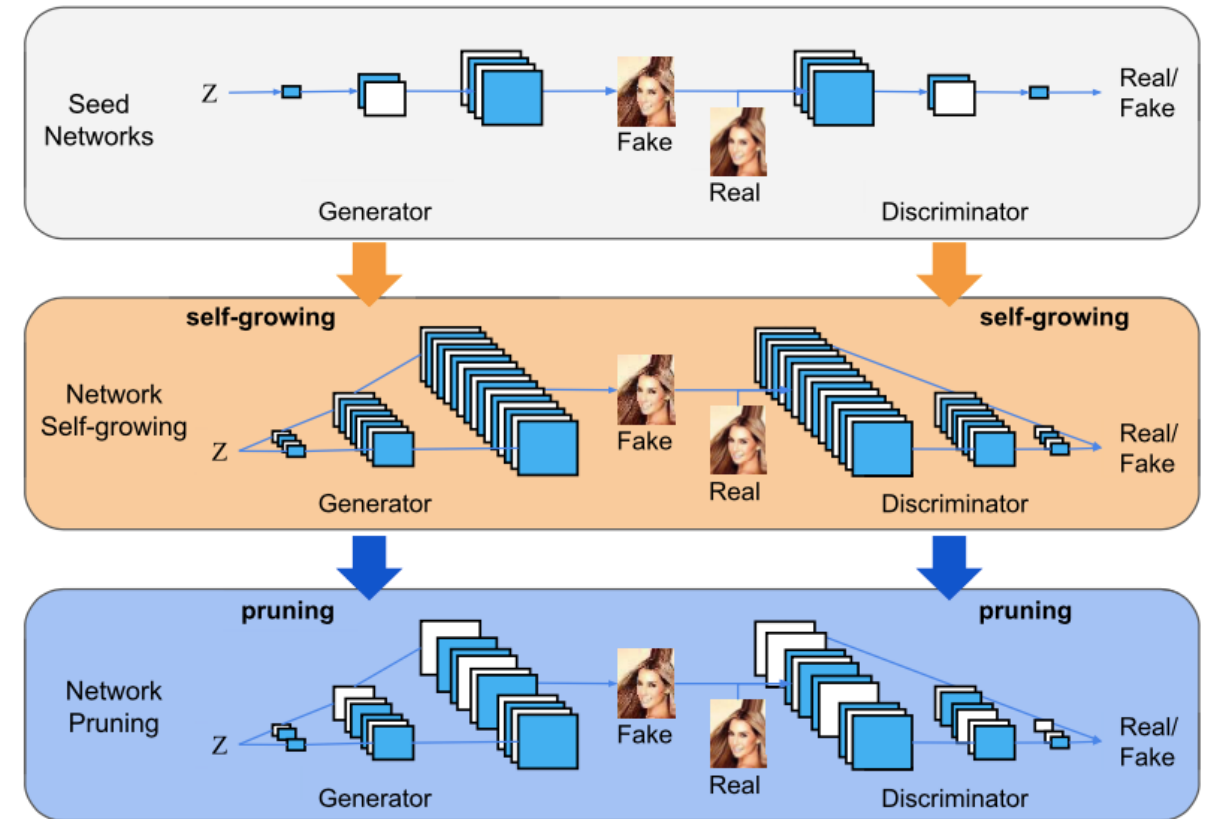
- ▶ Generator is a bitstream where each bit corresponds to a filter if the bit = 0 then the filter is pruned.
- ▶ Fitness:
 - the size of the network,
 - the compression distance,
 - the cycle loss.



H. Shu, ICCV 2019

Pruning during training

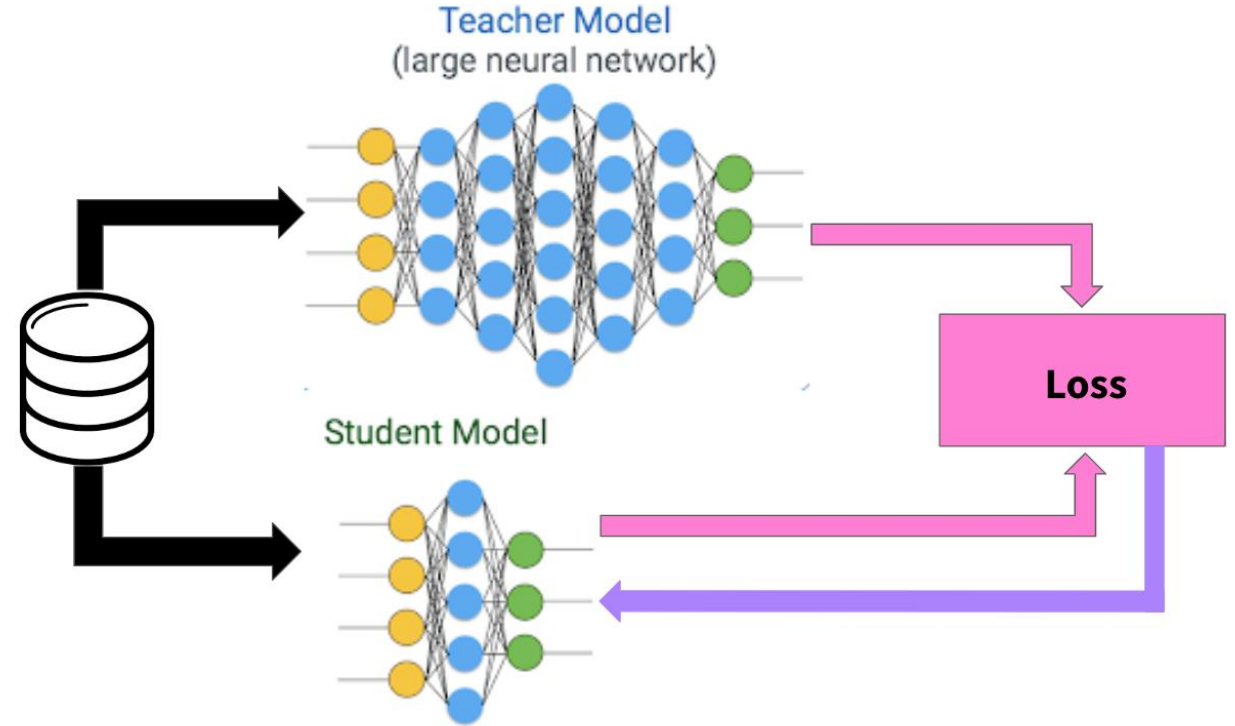
- ▶ Start from light-weight seed network.
- ▶ Grow width.
- ▶ Remove similar filters.



X. Song, SP-GAN: Self-Growing and Pruning Generative Adversarial Networks, IEEE TNN 2021

Knowledge Distillation Choices

- ▶ Teacher model acquiring
- ▶ Student model reconstruction
 - Pruning/architecture search/progressive growth
- ▶ Training Architecture
 - G/G+D/G+D+NN
- ▶ Loss functions

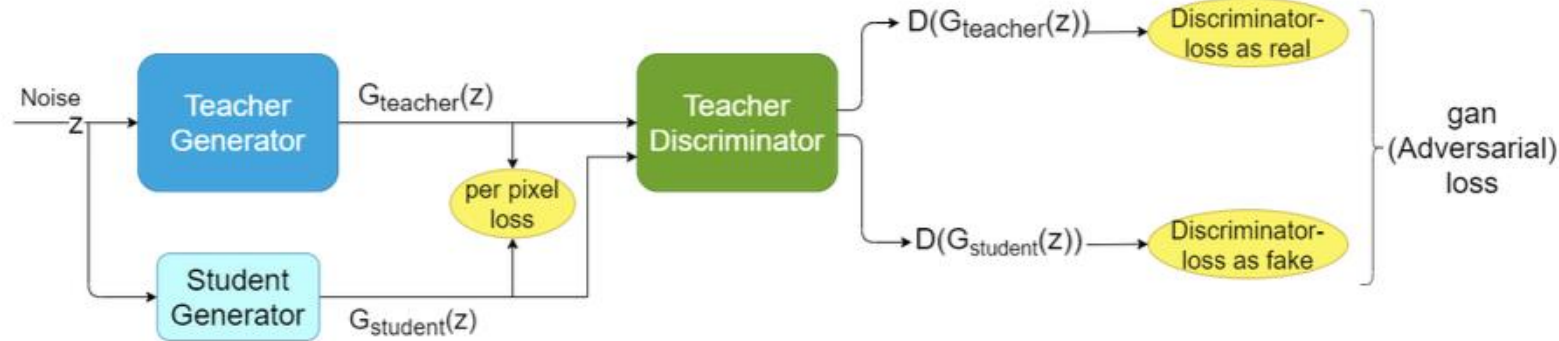


Direct Distillation

- Use pretrained teacher.
- Need per-pixel loss to stabilize student.

$$\mathcal{L}_{per_pixel} = loss(G_{student}(z), G_{teacher}(z))$$

$$\mathcal{L}_{recon} = \mathcal{L}_{gan} + \lambda \mathcal{L}_{per_pixel}$$

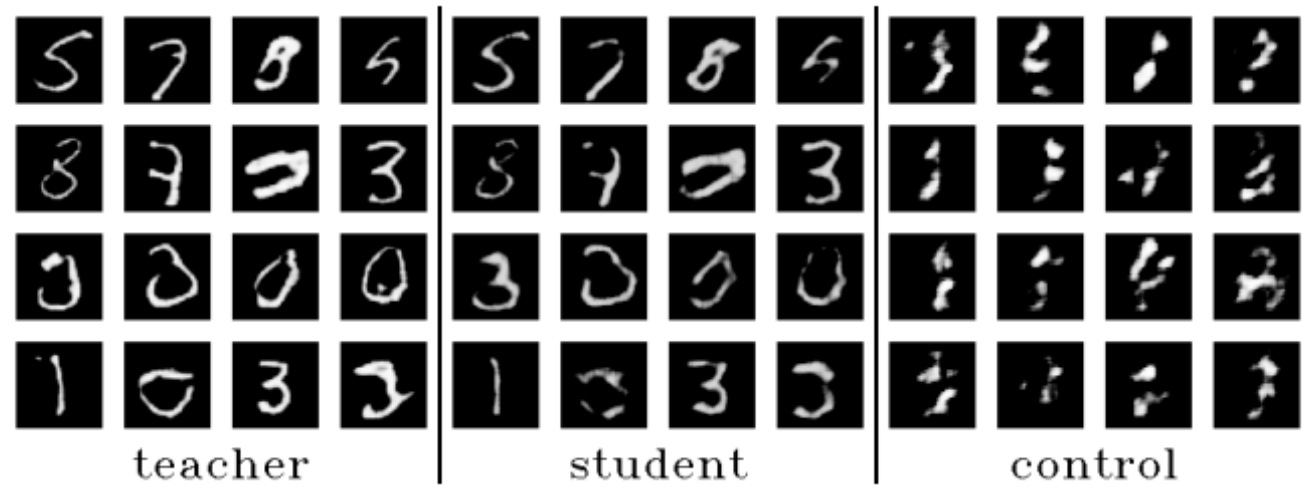
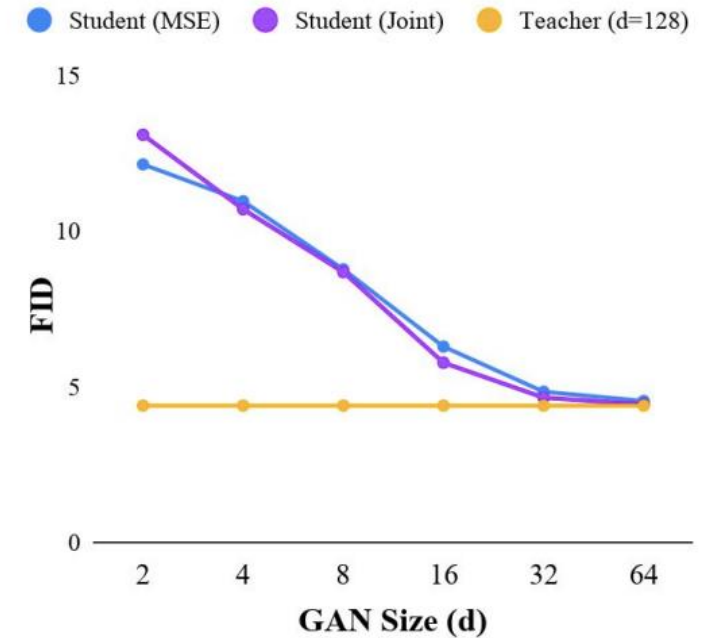


[Aguinaldo et al., arXiv:1902.00159](#)

Direct Distillation

- ▶ Clear dependence of quality on the complexity of student.
- ▶ Almost independent on loss-type.
- ▶ Results are much better than for direct training of small generator.

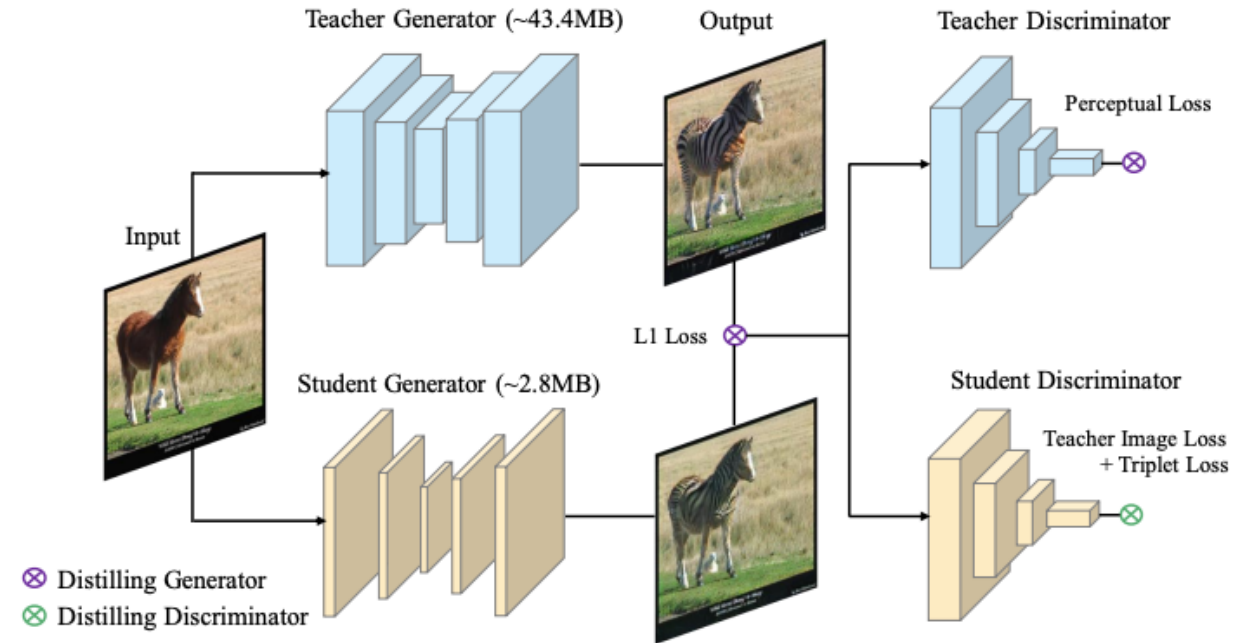
[Aguinaldo et al., arXiv:1902.00159](#)



Simultaneous Distillation

- ▶ Triplet loss: to consider the fact that student is closer to teacher than teacher to reality.

$$\mathcal{L}_{tri}(D_S) = \frac{1}{n} \sum_{i=1}^n \left[\|\hat{D}_S(y_i) - \hat{D}_S(G_T(x_i))\|_1 - \|\hat{D}_S(y_i) - \hat{D}_S(G_S(x_i))\|_1 + \alpha \right]_+$$



H. Chen et al., Distilling portable Generative Adversarial Networks for Image Translation, AAAI 2020

Simultaneous Distillation



(a) Student GANs with 1/2 channels of the teacher GAN.

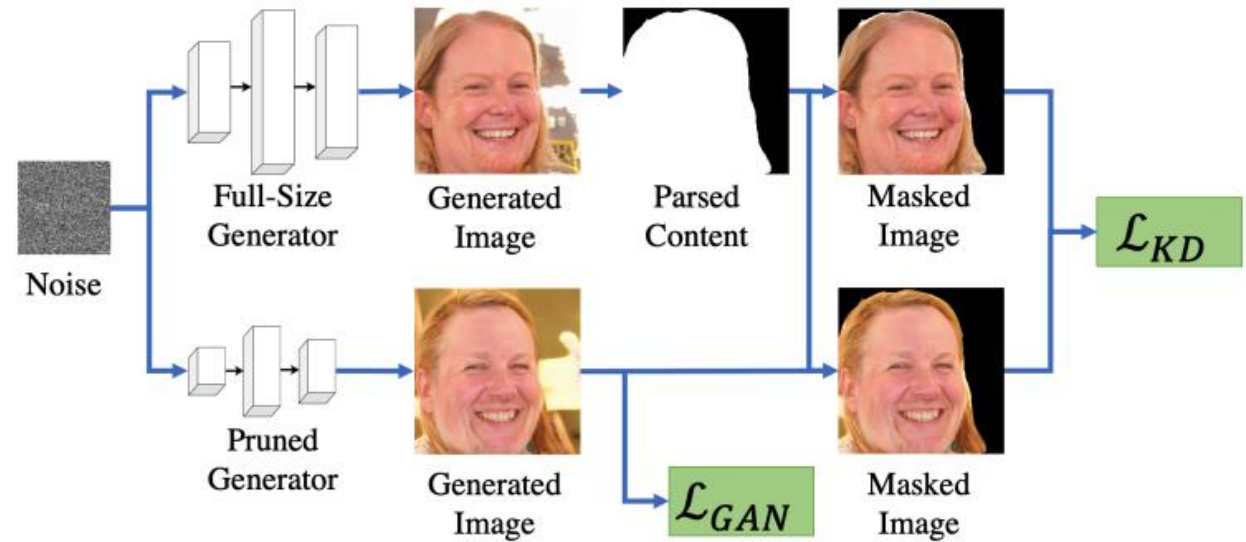


(b) Student GANs with 1/4 channels of the teacher GAN.

H. Chen et al., Distilling portable Generative Adversarial Networks for Image Translation, AAAI 2020

Content-aware loss

- ▶ Use masks to put correct attention to the content.
- ▶ The masks are specifically predefined.



Y. Liu et al., CVPR2021

Outlook

- ▶ More methods are available (like quantization).
- ▶ Current state-of-the-art methods use loss with several components.
- ▶ No unified approach exist.